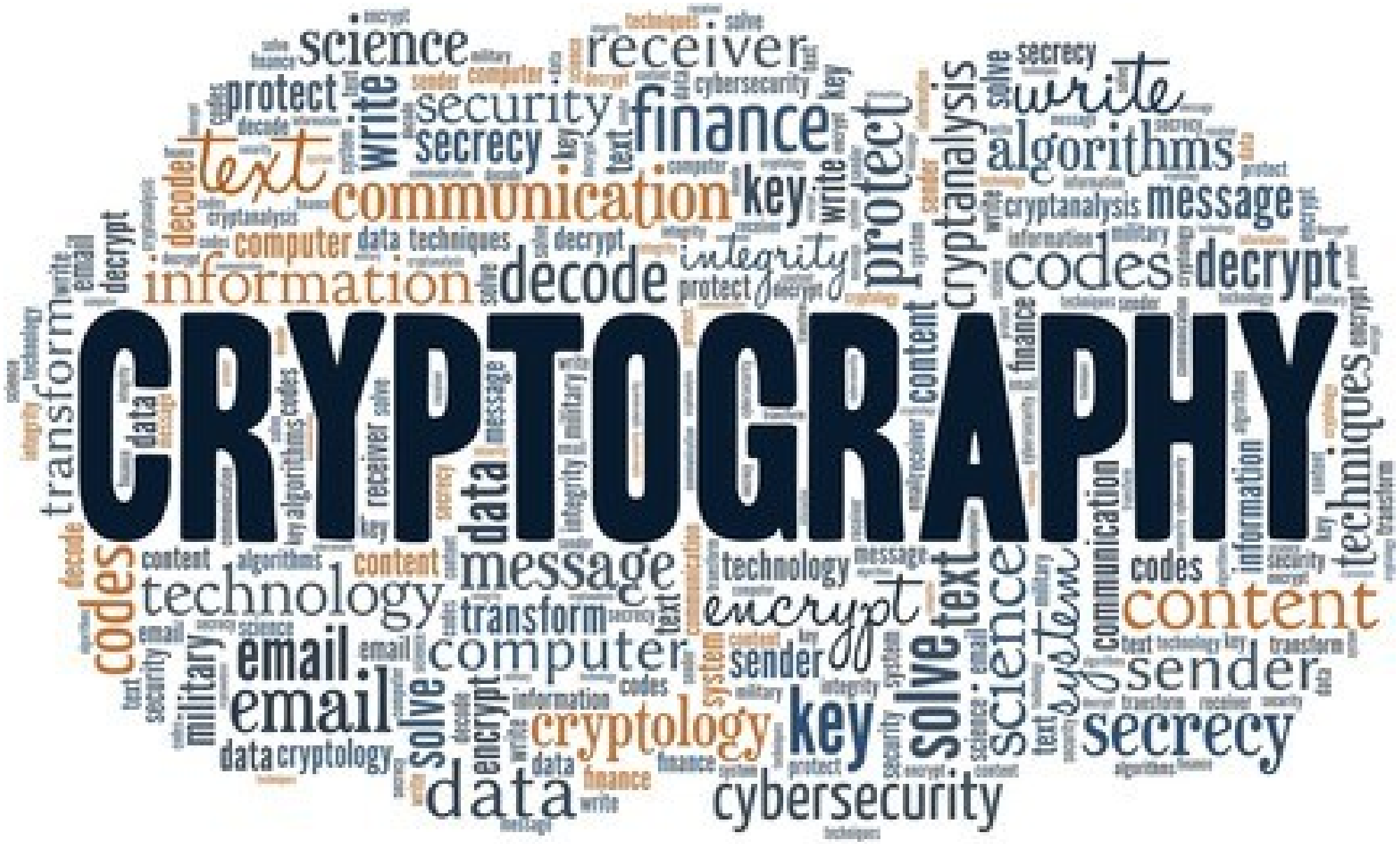

CRYPTOGRAPHY
science
receiver
protect
write
security
finance
secrecy
text
communication
key
algorithms
message
cryptanalysis
computer
data
techniques
decrypt
integrity
decode
information
codes
content
technology
encrypt
transform
email
cryptology
solve
sender
system
military
cybersecurity

Applied Cryptography

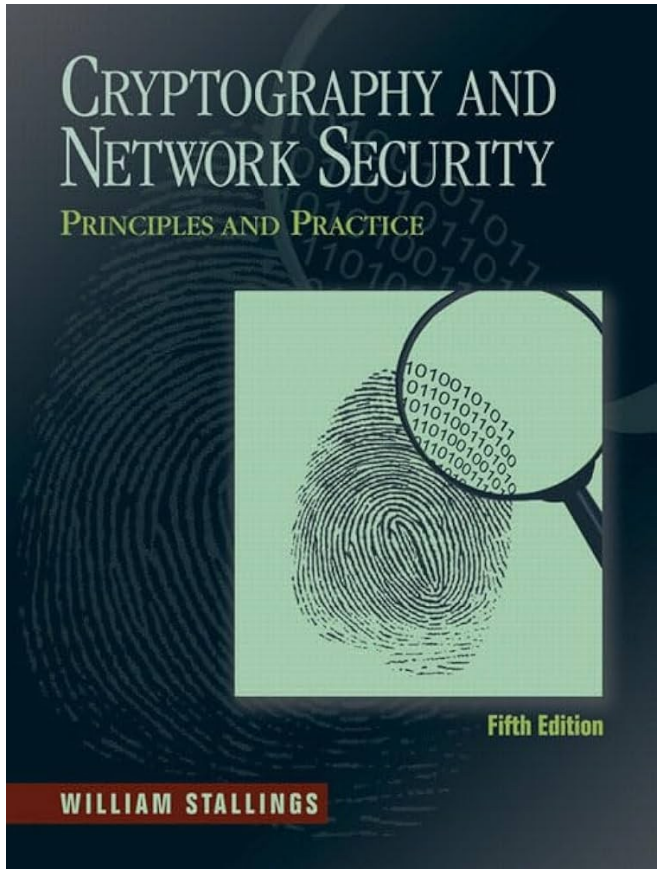
Dr. Hanan Hindy

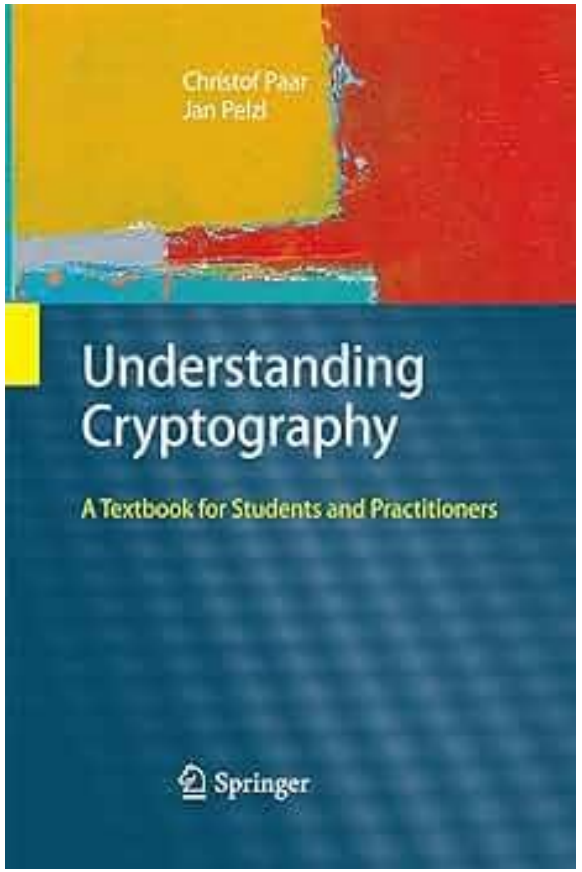
hanan.hindy@cis.asu.edu.eg




Expectations

Main Textbooks





Grading Scheme

- Midterm 15%
- Quizzes 15%
- Project 20%

- Final 50%

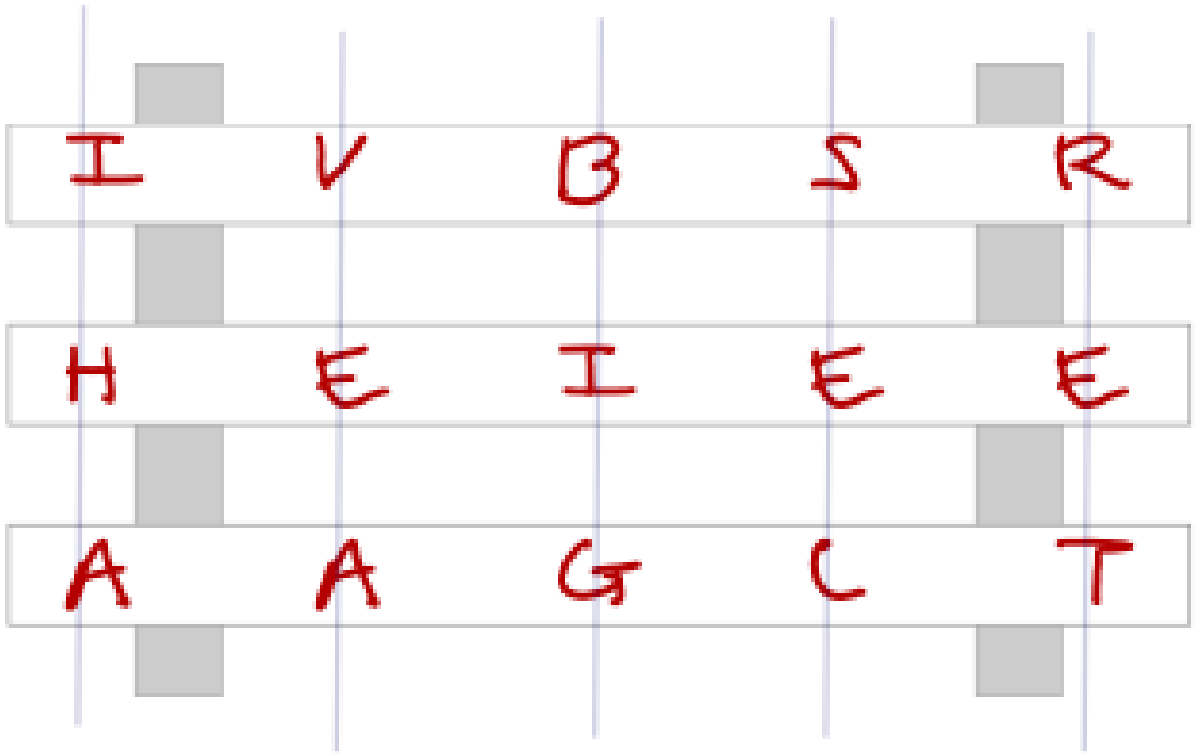
Office Hours

- TBA
- Email: hanan.hindy@cis.asu.edu.eg

Proposed Course Outline

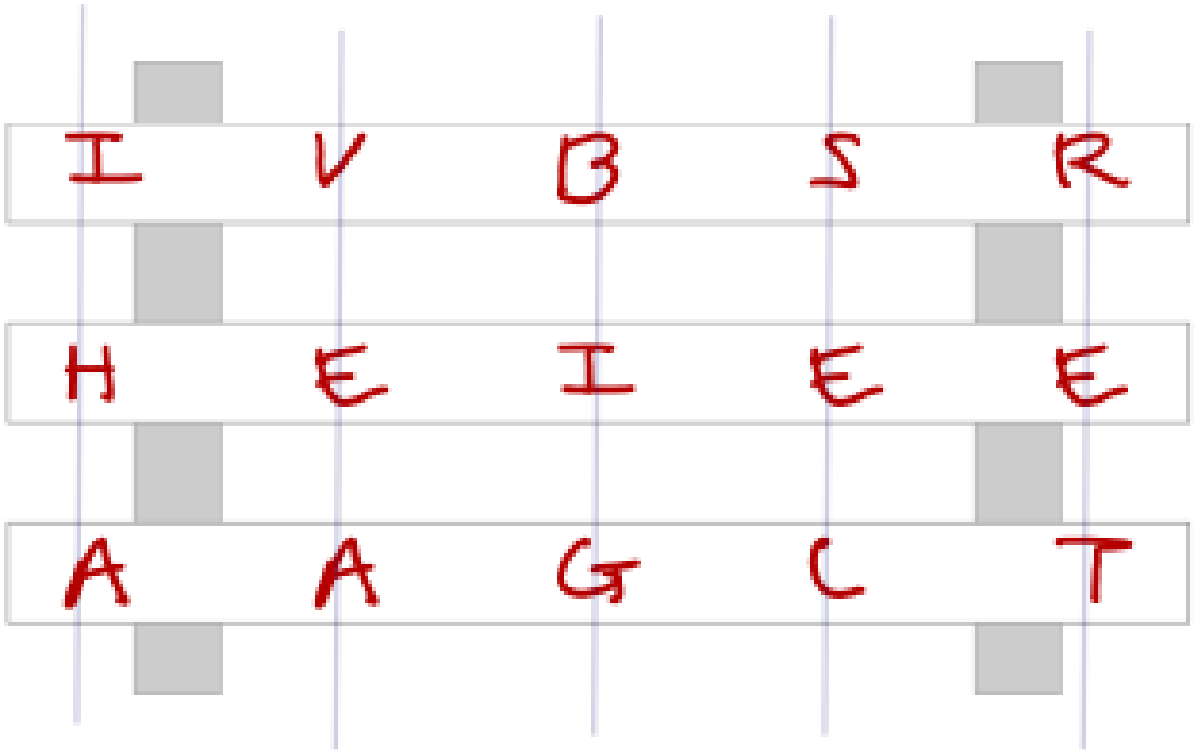
Introduction to Cryptography: - Security model for Information Transfer - Basic terminologies: encryption, decryption, plaintext, ciphertext, keys - Types of cryptography: Symmetric vs. Asymmetric cryptography - Classical Techniques: o Ceaser   o Monoalphabetic
Classical Techniques Cont: o Polyalphabetic   o Playfair   o Columnar   o Rail Fence
Data Encryption Standard Modes of Operation
Finite Fields (Math Foundation)
Advanced Encryption Standard
Number Theory (prime numbers, random no, etc) RC4
Asymmetric Key Cryptography RSA algorithm
Diffie-Hellman Key Management and Distribution Certificates
Cryptographic Hash Functions Password Management
Quantum Cryptography

Let's Play..

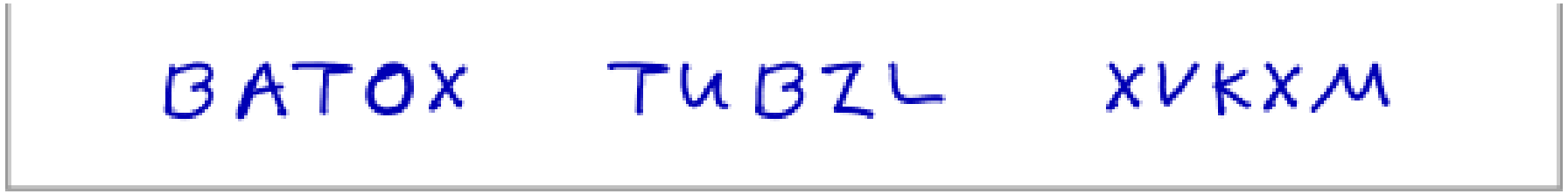


Let's Play..

I HAVE A BIG SECRET



Let's Play..



Let's Play..

BATOX TUBZL XVKXM

IHAVE ABIGS ECRET

BATOX TUBZL XVKXM

I HAVE A BIG SECRET

Cdzq Rstcdmsr,

Sghr vddj gzr addm z khsskd cheedqdms eqnl zkk sgd nsgdqr xnt’ud dwodqhdmbdc adenqd. Hs gzr addm xntq kzrs ehqrs vddj ne tmhudqrhsx -- z ogqzrd sgzs bzqqhdr ansg oqhcd zmc mnrszkfhz.

Xntq kzrs ehqrs vddj hr z qdlhmcdq sgzs dudqx kdbstqd xnt zssdmc mnv, dudqx oqnidbs xnt vnqj nm, dudqx kzsd-mhfgs rstcx rdrrhnm, hr mn knmfdq itrs zants ozrrhmf z bntqrd. Hs’r zants bnlokdshmf z rsnqx xnt’ud addm vqhshmf enq xdzqr. Hs’r zants oqdozqhmf sn rsdo hmsn sgd vnqkc adxnmc sgdrd vzkkr, bzqqxhmf vhsg xnt sgd kdrrnmr zmc sgd dwodqhdmbdr sgzs rgzodc xnt.

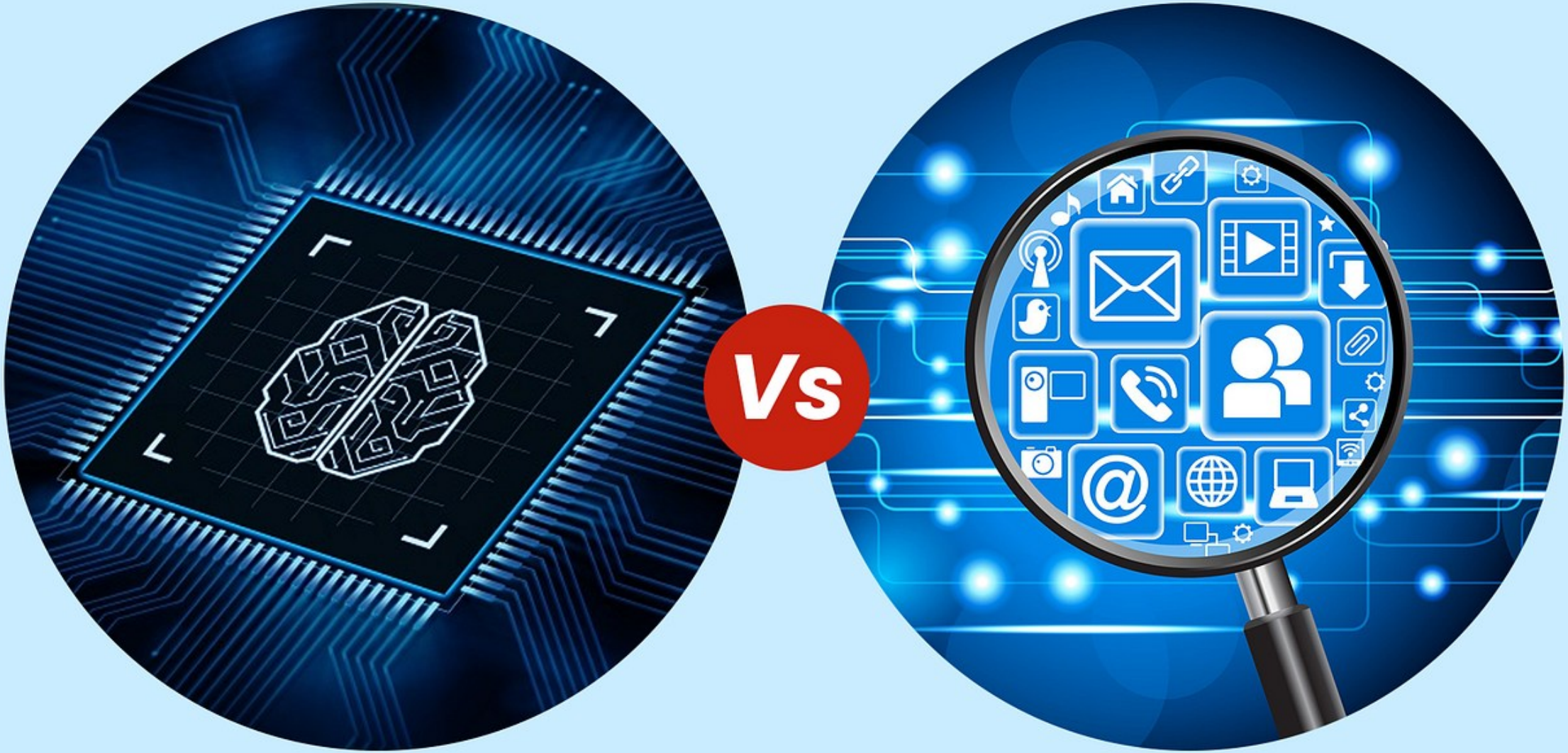
Cnm’s qtrg sgqntfg sghr ehmzk xdzq -- khud hs etkkx. Zooqdbhzsd sgd rlzkk lnldmsr: sgd kztfgsdq vhsg xntq oddqr, sgd zcuhbd eqnl xntq oqnedrrnqr, sgd qntshmd ne bzlotr khed sgzs xnt lzx nmd czx knnj azbj nm enmckx.

Lzjd sghr xdzq bntms.

Information Security and Cryptography

Information Security

- Information Security is the **protection** of information from **unauthorized** **access**, **usage**, **disclosure**, **modification**, or any **misuse**.



Data

Information

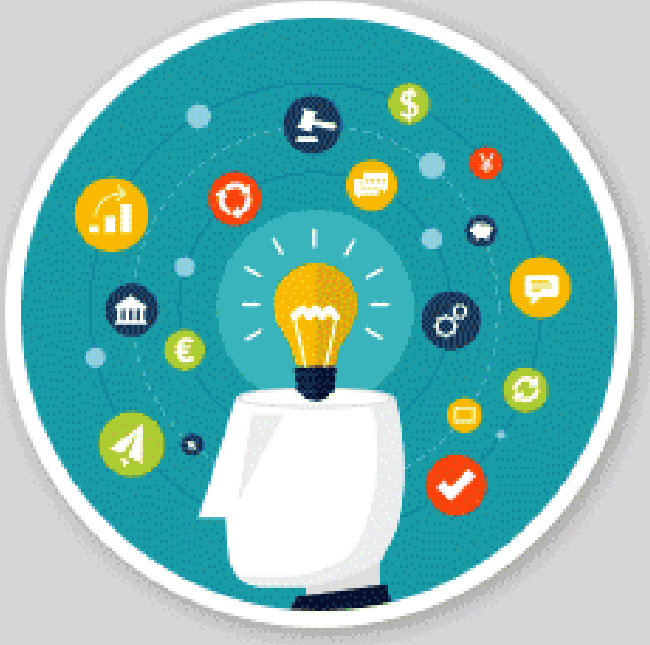
DIFFERENCE BETWEEN DATA AND INFORMATION

BYJU'S The Learning App



DATA

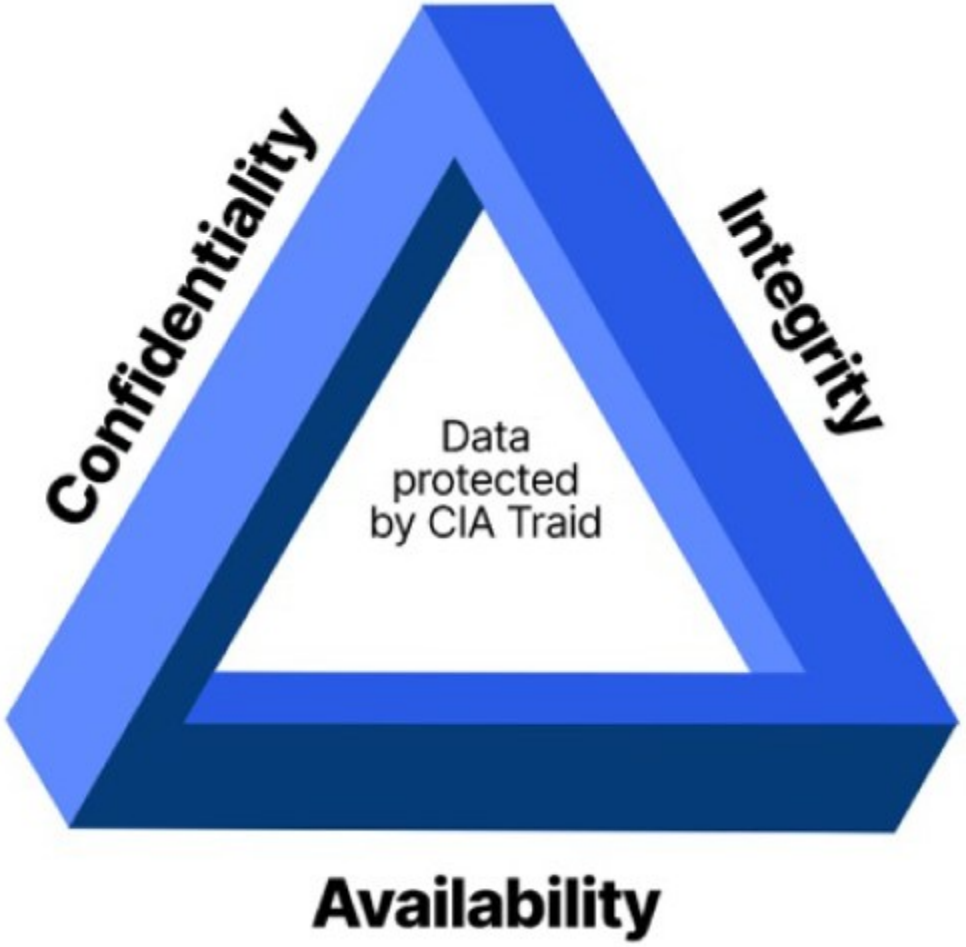
Data is raw, unorganized facts that need to be processed. Data can be something simple and seemingly random and useless until it is organized.



INFORMATION

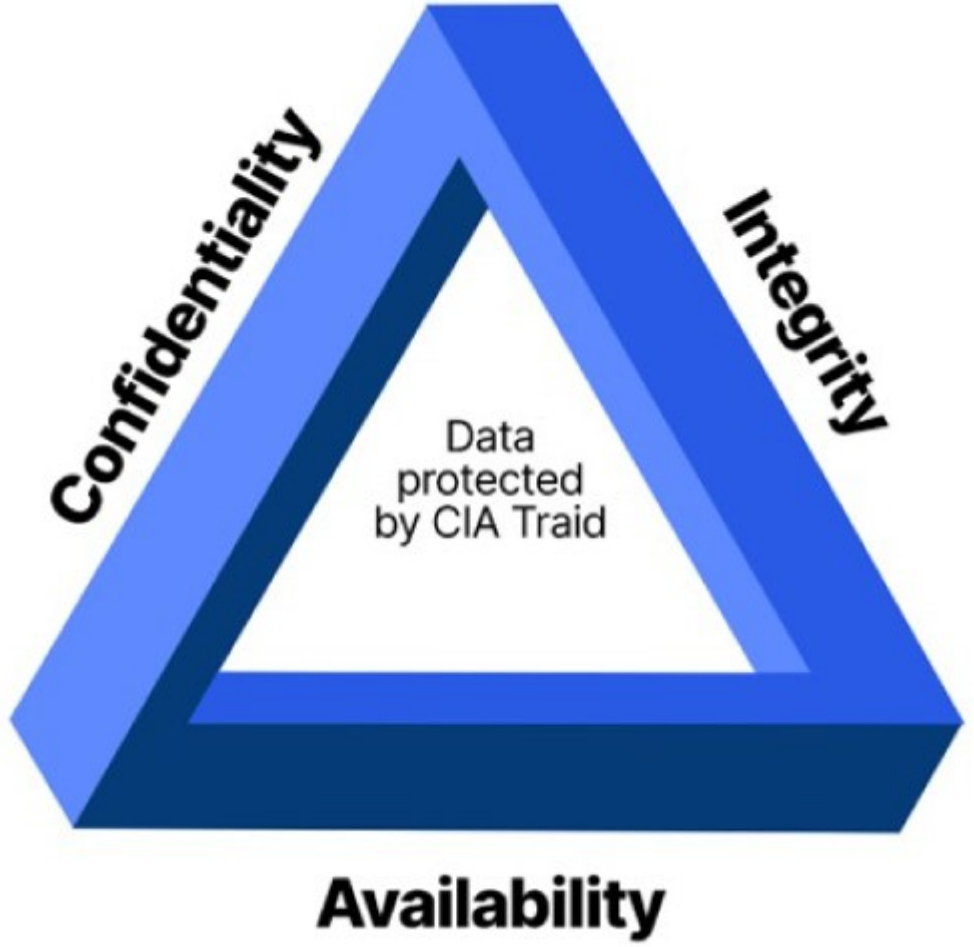
When data is processed, organized, structured or presented in a given context so as to make it useful, it is called information.

Principles of Information Security



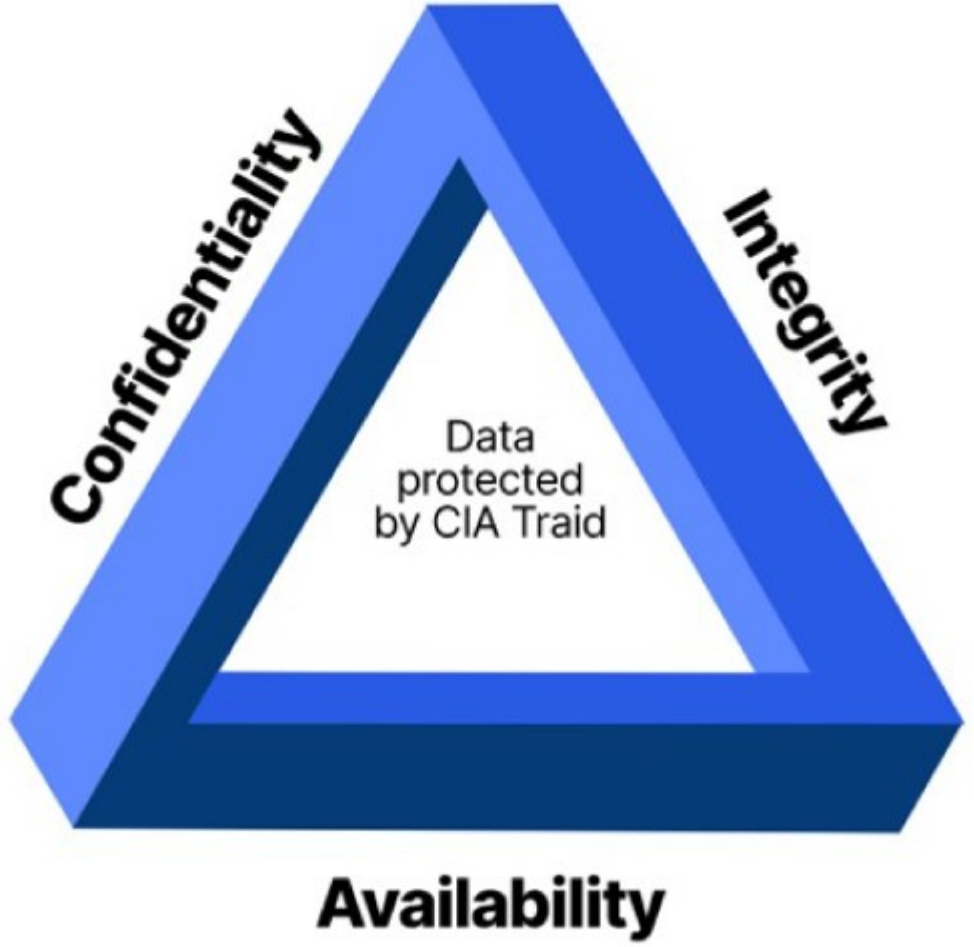
Principles of Information Security

- **Confidentiality** evaluates the protection from **unofficial** information broadcasting.
- The goal of the confidentiality principle is to **keep sensitive information private** and to ensure that it is manifest and available only to those who are **authorized** to use it to fulfill the important or institutional projects.



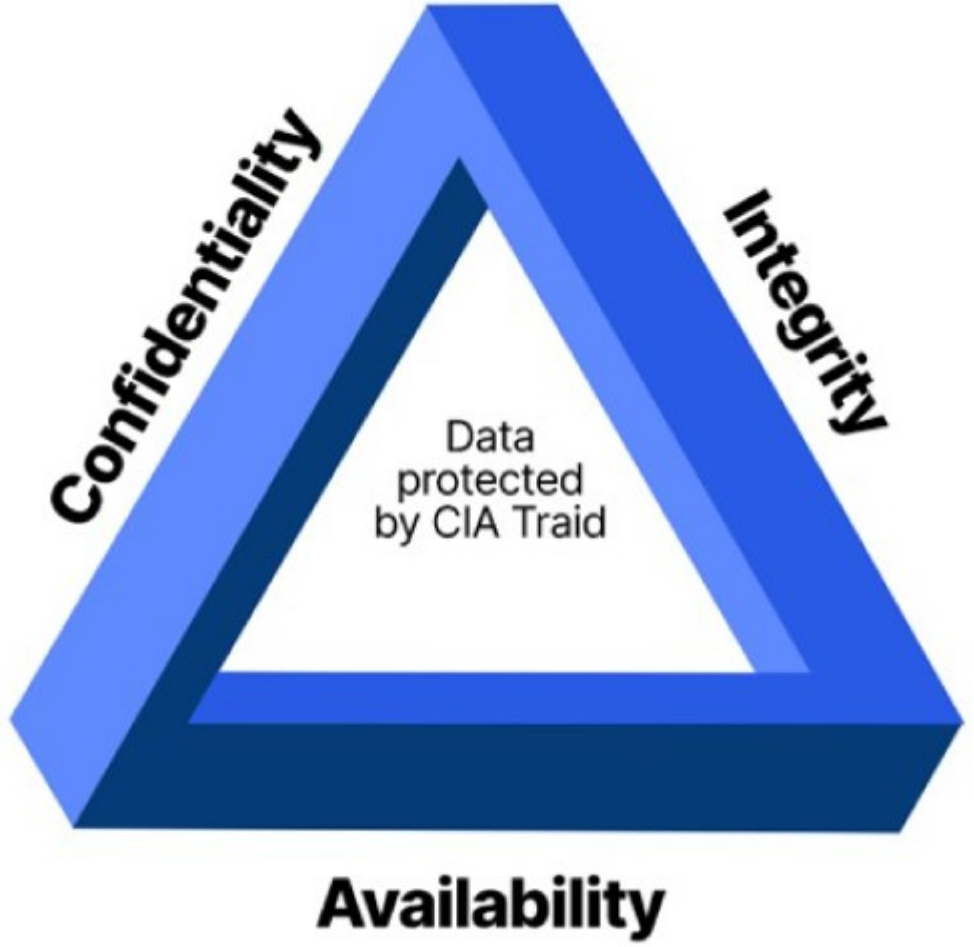
Principles of Information Security

- The main purpose of the **integrity** principle is to protect data from **being modified in any uncertified way**.
- It provides stability and guarantees that data is **correct**, **authentic**, and **not modified** (addition, deletion, etc.).
- It continuously protects data from modification, whether coincidentally or unfaithfully.



Principles of Information Security

- The primary objective of **availability** is to verify that the complete data is **available** every time (or at any moment) **whenever** an **official** person needs it.
- This means availability is the safeguard to a system's capacity to build technology more effectively, software tools, applications, and data accessible when required for any institutional tasks or any institutional worker.



Information Security (NIST)

The protection of information and information systems from access, use, disclosure, unauthorized disruption, modification, or destruction in order to provide **confidentiality**, **integrity**, and **availability.**

Information Security Objectives

Privacy or confidentiality	Keeping information secret from all but those who are authorized to see it.
Data integrity	Ensuring information has not been altered by unauthorized or unknown means.
Entity authentication or identification	Corroboration of the identity of an entity (e.g., a person, a computer terminal, a credit card, etc.).
Message authentication	Corroborating the source of information; also known as data origin authentication.
Signature	A means to bind information to an entity authorization
Validation	A means to provide timeliness of authorization to use or manipulate information.
Access control	Restricting access to resources to privileged entities.
Certification	Endorsement of information by a trusted entity.
Time stamping	Recording the time of creation or existence of information.
Witnessing	Verifying the creation or existence of information by an entity other than the creator.
Receipt	Acknowledgement that information has been received.
Confirmation	Acknowledgement that services have been provided.
Ownership	A means to provide an entity with the legal right to use or transfer a resource to others.
Anonymity	Concealing the identity of an entity involved in some process.
non-repudiation	Preventing the denial of previous commitments or actions.
Revocation	Retraction of certification or authorization.

Cryptographic Goals

Privacy or confidentiality	Keeping information secret from all but those who are authorized to see it.
Data integrity	Ensuring information has not been altered by unauthorized or unknown means.
Entity authentication or identification	Corroboration of the identity of an entity (e.g., a person, a computer terminal, a credit card, etc.).
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Revocation	Retraction of certification or authorization.

Cryptography…

- Cryptography (the art and science of secure communication) is often considered **a solution** to security.
- It can often form part of the solution, but in itself it is not a full solution to a problem.
- It can make security of systems stronger but also weaker if weak crypto solutions are being implemented.

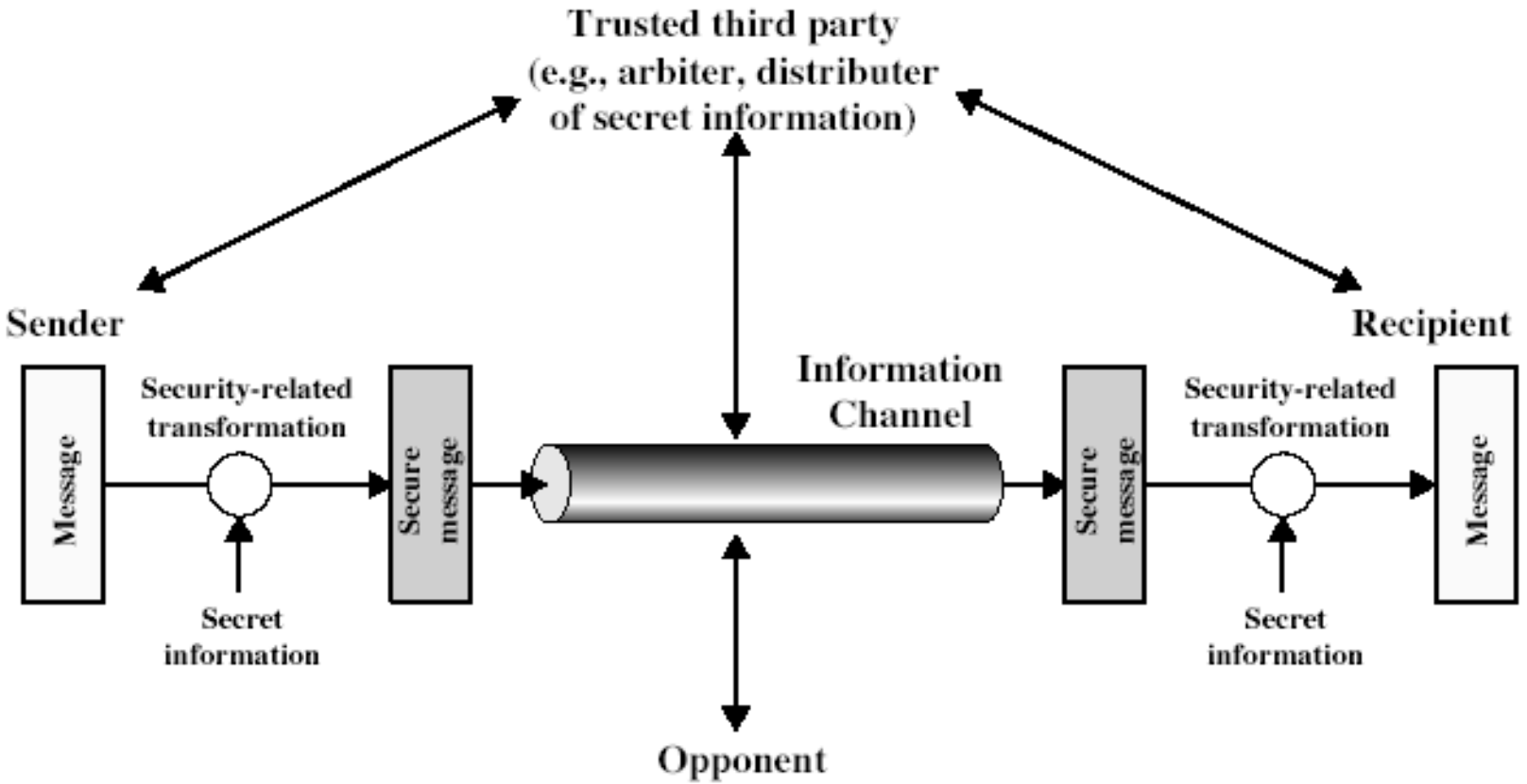
Never trust a cryptographic system that has not been analyzed by experts (and still then there is likely to be flaws!)

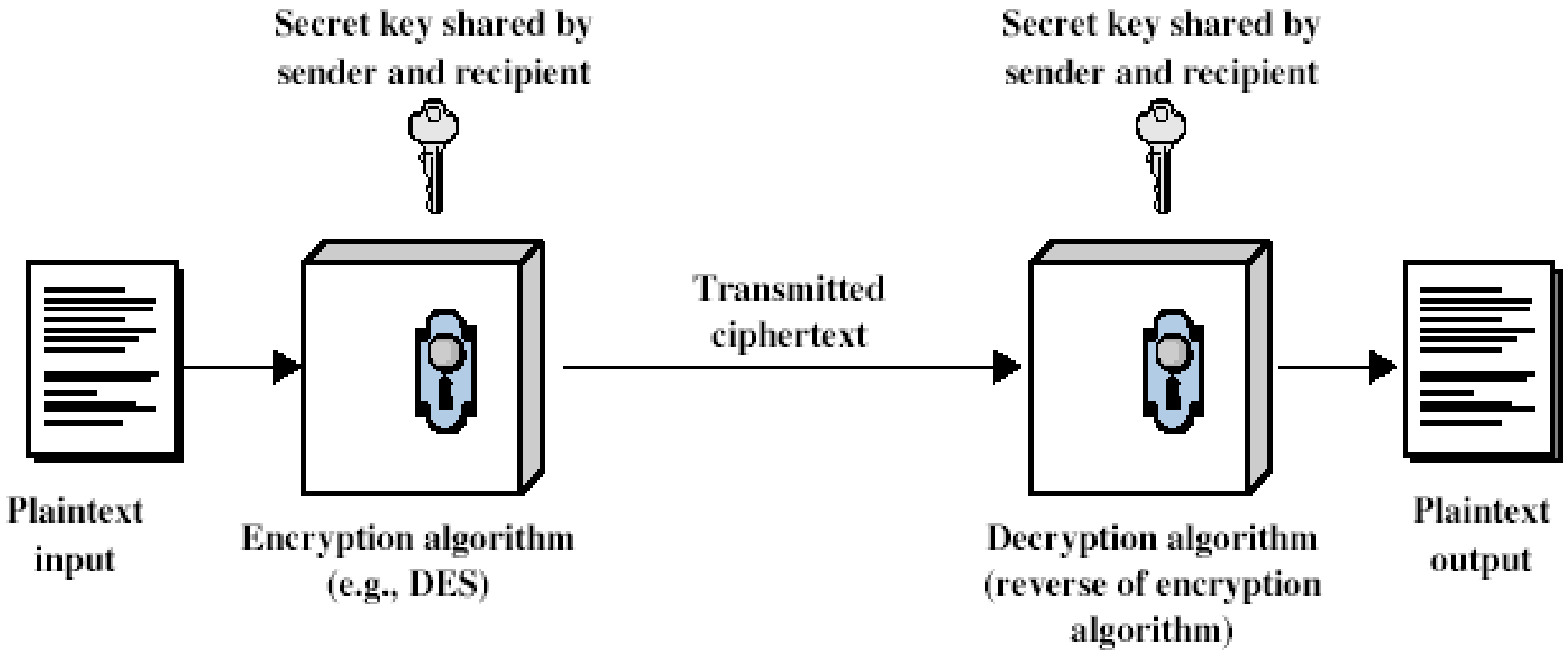
Cryptography

- Cryptography is the study of mathematical techniques related to aspects of information security such as confidentiality, data integrity, entity authentication, and data origin authentication.

Secret (crypto) Writing (graphy)

Security Model of Information Transfer



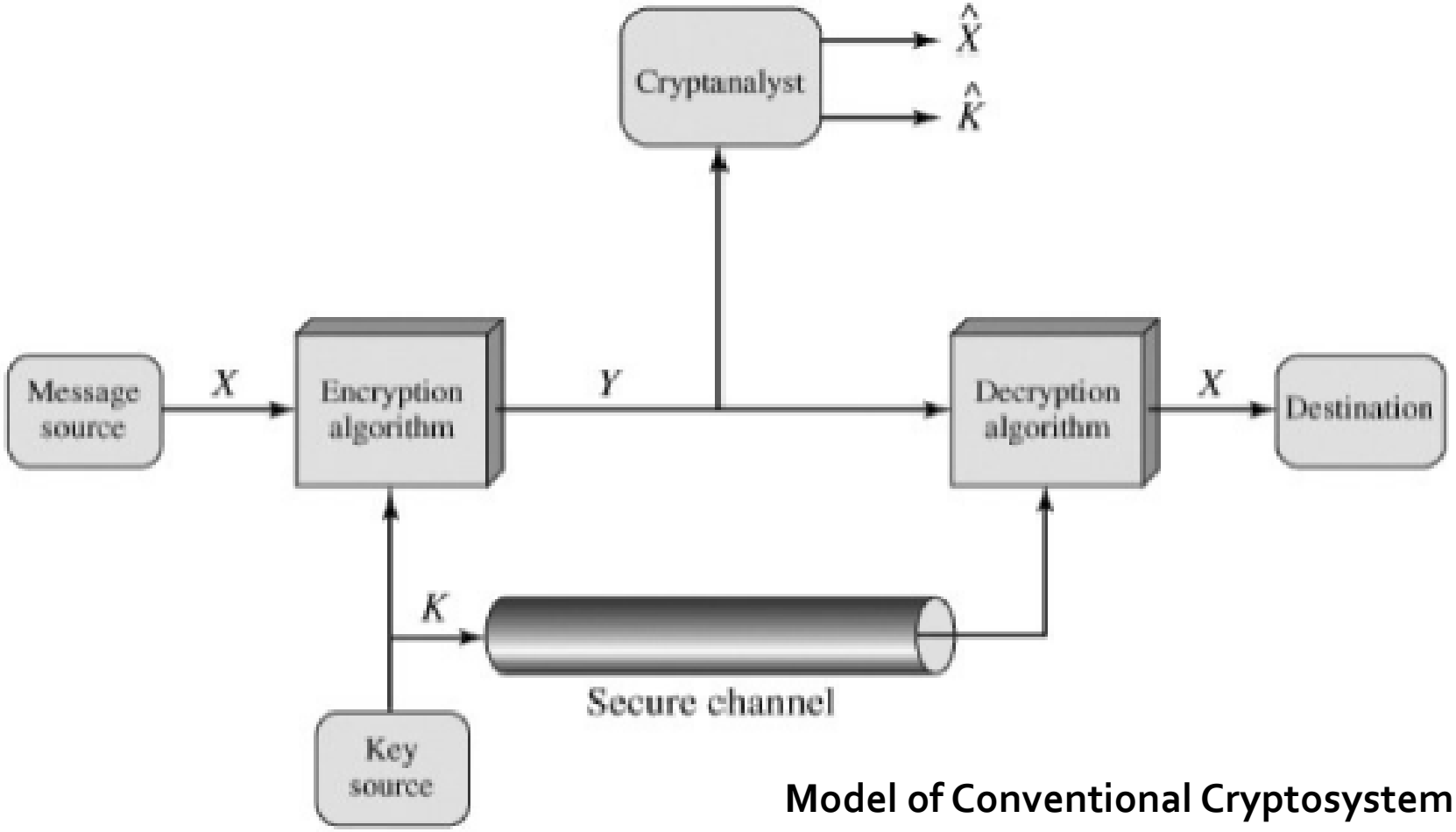


Simplified Model of Conventional Encryption

Terminologies

- **m** represents the original **plaintext** message.
- **c** represents the **ciphertext**, i.e. encrypted version of the message.
- **E** represents **encryption**: is the process of converting plaintext into ciphertext, especially to prevent unauthorized access.
- **D** represents **decryption**: is the reverse process, converting ciphertext back into the original plaintext.
- **k** represents **key**: is the key used by the encryption, decryption function
- **|k|** represents **keyspace**: is the number of possible keys

Classical Encryption Techniques

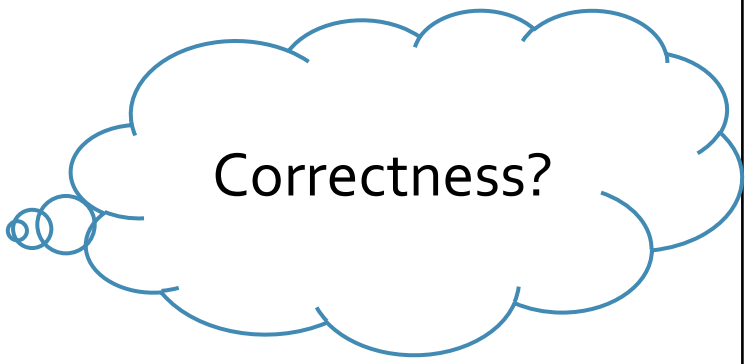


Model of Conventional Cryptosystem

Cryptosystem



- A cryptosystem is a five-tuple (P,C,K,E,D) where the following conditions are satisfied:
  - P is a finite set of possible plaintexts:
  - C is a finite set of possible ciphertexts:
  - K is the keyspace, is the a finite set of possible keys:
  - For each k ∈ K, there is an encryption rule $e_k \in E$ and a corresponding decryption rule $d_k \in D$. Where
    - $e_k : P \rightarrow C$ and
    - $d_k : C \rightarrow P$
      - are functions such that for all M∈P, we have $\mathbf{d_k(e_k(M)) = M}$



Kerckhoffs' principle

1° Le système doit être matériellement, sinon mathématiquement, indéchiffrable ;

2° Il faut qu'il n'exige pas le secret, et qu'il puisse sans inconvénient tomber entre les mains de l'ennemi ;

3° La clef doit pouvoir en être communiquée et retenue sans le secours de notes écrites, et être changée ou modifiée au gré des correspondants ;

4° Il faut qu'il soit applicable à la correspondance télégraphique ;

5° Il faut qu'il soit portatif, et que son maniement ou son fonctionnement n'exige pas le concours de plusieurs personnes ;

6° Enfin, il est nécessaire, vu les circonstances qui en commandent l'application, que le système soit d'un usage facile, ne demandant ni tension d'esprit, ni la connaissance d'une longue série de règles à observer.

Kerckhoffs' principle

- Kerckhoffs' principle is a fundamental concept in cryptography. It states that the security of a cryptographic system shouldn't rely on the secrecy of the algorithm.
- Instead, it should be based on the secrecy of the cryptographic key.
- A good cryptographic system should remain secure even if the algorithm used is known.



Auguste Kerckhoffs (1883):
The enemy knows the system

The cipher should remain secure even if **the adversary knows the specification of the cipher.**

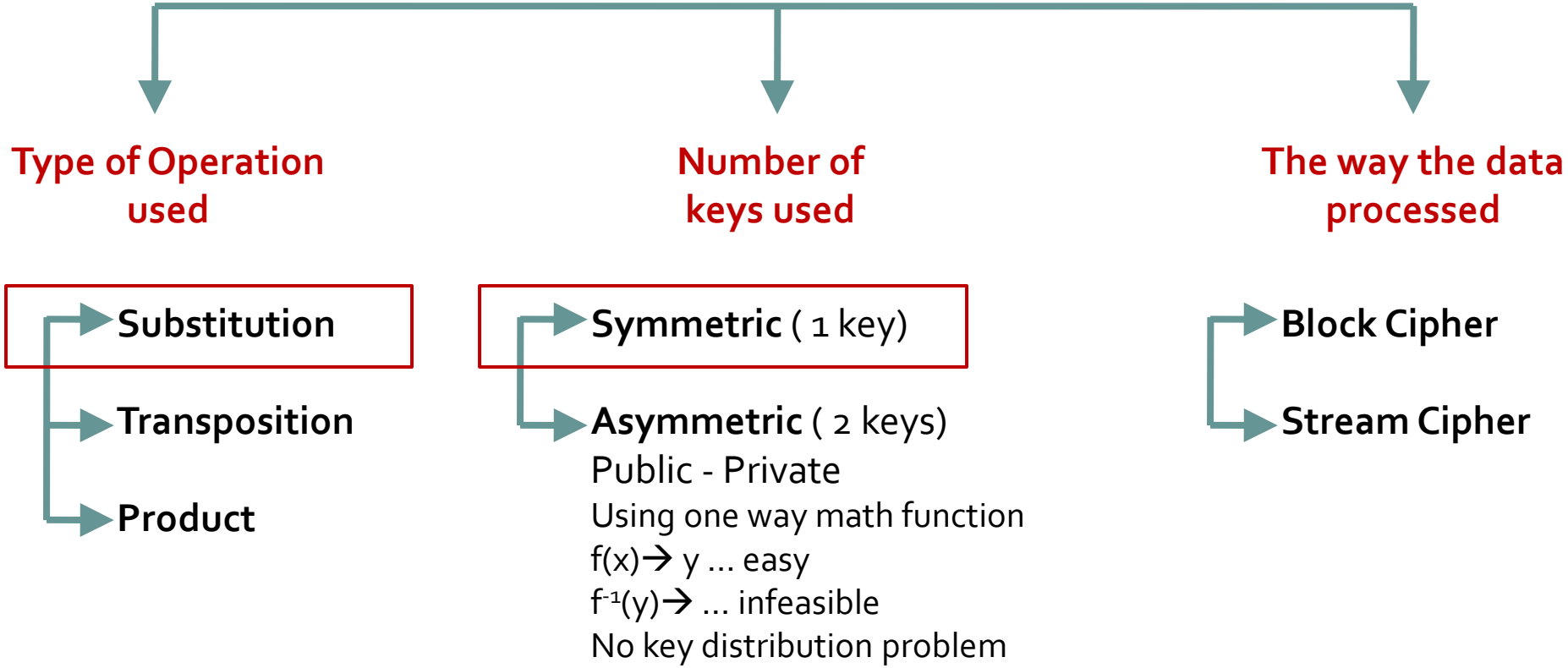
The only thing that is **secret** is a

short key **k**

that is **usually chosen uniformly at random**

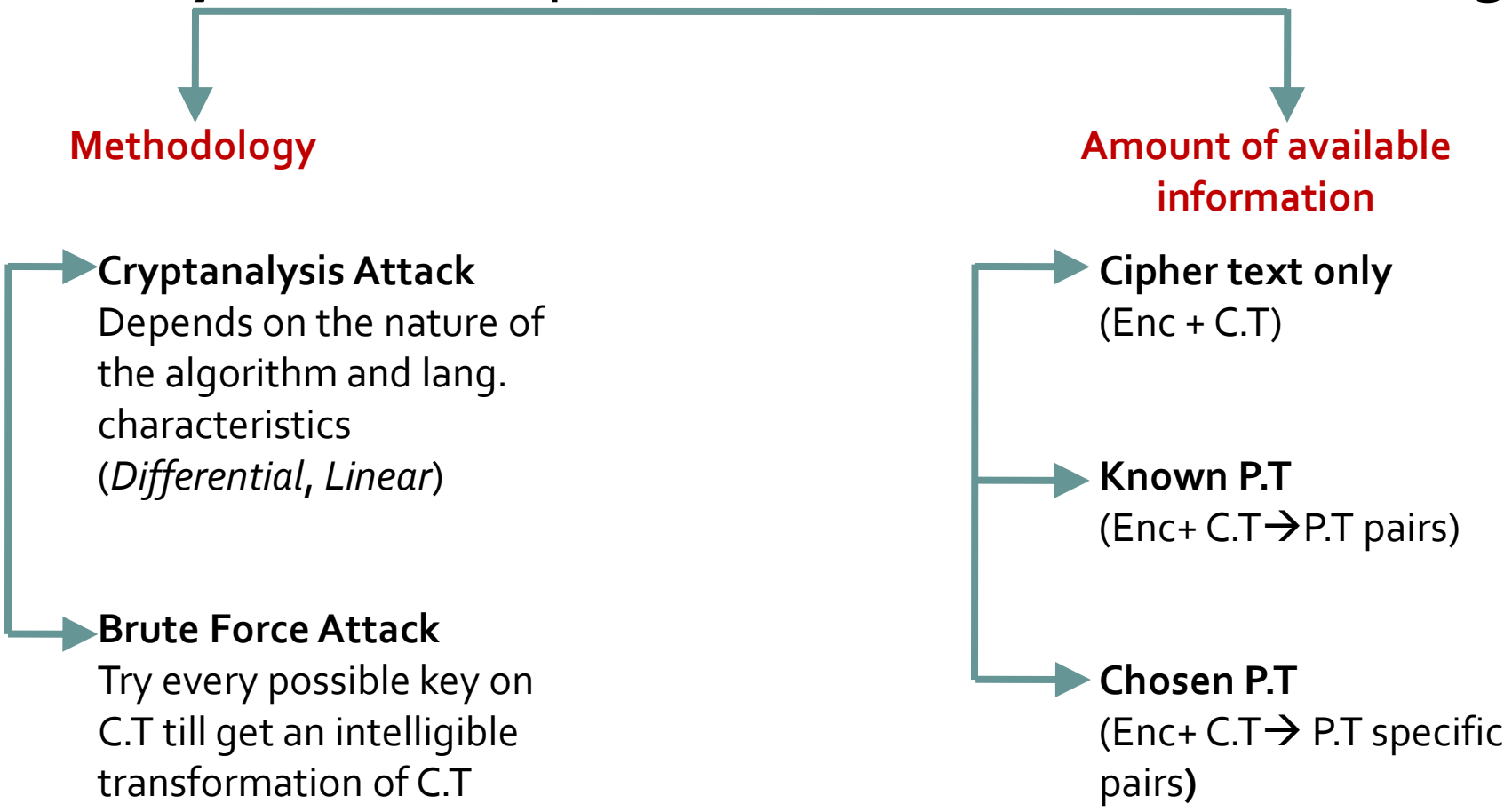
Classical Encryption Techniques

Encryption techniques can be classified according to:



Classical Encryption Techniques

Cryptanalysis techniques can be classified according to:



Security Encryption Schemes

- An encryption scheme is **unconditionally secure** if the ciphertext generated by the scheme does not contain enough information to determine uniquely the corresponding plaintext, no matter how much ciphertext is available.

Security Encryption Schemes

- An encryption scheme is said to be **computationally secure** if:
  - The cost of breaking the cipher exceeds the value of the encrypted information
  - The time required to break the cipher exceeds the useful lifetime of the information

Classical Substitution Ciphers

- Where letters of plaintext are **replaced** by other letters or by numbers or symbols
- Or if plaintext is viewed as a sequence of bits, then **substitution** involves replacing plaintext bit patterns with ciphertext bit patterns

Substitution Techniques

- Ceaser
- Monoalphabetic
- Polyalphabetic
- Playfair
- Hill Cipher

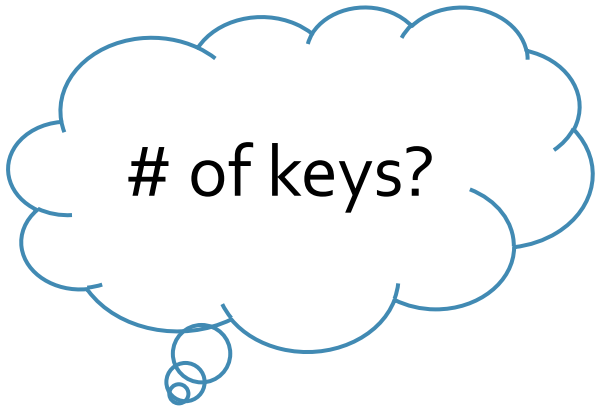
1) Caesar Cipher

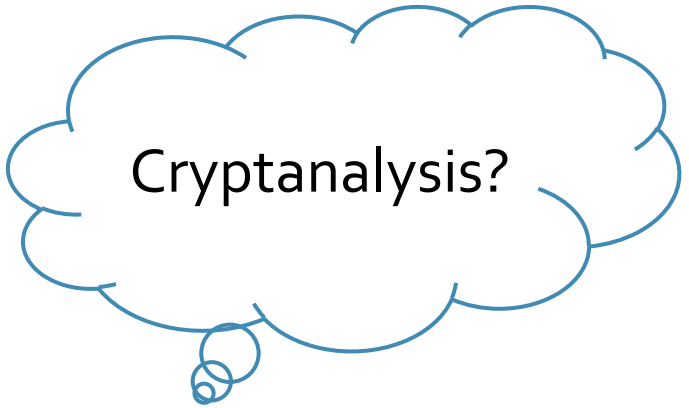
- Earliest known substitution cipher (invented by Julius Caesar)
- First attested use in military affairs
- Replaces each letter by 3rd letter down in the alphabet.
- Example:
  - Meet me after the toga party
  - PHHW PH DIWHU WKH WRJD SDUWB
- Mathematically give each letter a number
- Then have Caesar cipher as:
  - $c = E(p) = (p + k) \bmod (26)$
  - $p = D(c) = (c - k) \bmod (26)$

A	0
B	1
C	2
D	3
E	4
F	5
G	6
H	7
I	8
J	9
K	10
L	11
M	12
N	13
O	14
P	15
Q	16
R	17
S	18
T	19
U	20
V	21
W	22
X	23
Y	24
Z	25

1) Caesar Cipher Example

- P.T = computer, Key = 8
- C.T?





A	0
B	1
C	2
D	3
E	4
F	5
G	6
H	7
I	8
J	9
K	10
L	11
M	12
N	13
O	14
P	15
Q	16
R	17
S	18
T	19
U	20
V	21
W	22
X	23
Y	24
Z	25

1) Caesar Cipher Example

- P.T = computer, Key = 8
- C.T = KWUXCBMZ

A	0
B	1
C	2
D	3
E	4
F	5
G	6
H	7
I	8
J	9
K	10
L	11
M	12
N	13
O	14
P	15
Q	16
R	17
S	18
T	19
U	20
V	21
W	22
X	23
Y	24
Z	25

Cryptanalysis

```
rafel@Rafaels-iMac python_code %
1 --> dtwvghqtegcvvcem
2 --> csvufgpsdfbuubdl
3 --> bruteforceattack
4 --> aqtsdenqbdzsszbj
5 --> zpsrcdmpacyrryai
6 --> yorqbclozbxqqxzh
7 --> xnqpabknyawppwyg
8 --> wmpozajmxzvoovxf
9 --> vlonyzilwyunnuwe
10 --> uknmxyhkvxtmmtvd
11 --> tjmlwxgjuwsllsuc
12 --> silkvwfitvrkkrtb
13 --> rhkjuvehsuqjjqsa
14 --> qgjitudgrtpiiprz
15 --> pfihstcfqsohhoqy
16 --> oehgrsbeprnggnpx
17 --> ndgfqradoqmffmow
18 --> mcfepqzcnpleelnv
19 --> lbedopybmokddkmu
20 --> kadcnoxalnjccjlt
21 --> jzcbmnwzkmibbiks
22 --> iybalmvyjlhaahjr
23 --> hxazkluxikgzzgiq
24 --> gwzyjktwhjfyyfhp
25 --> fvyxijsvgiexxego
rafel@Rafaels-iMac python_code %
```

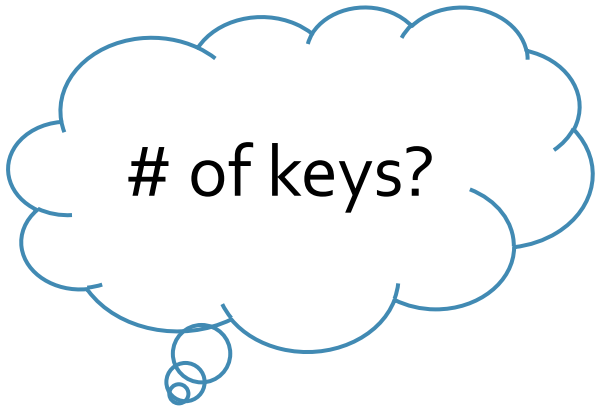
2) Monoalphabetic Cipher

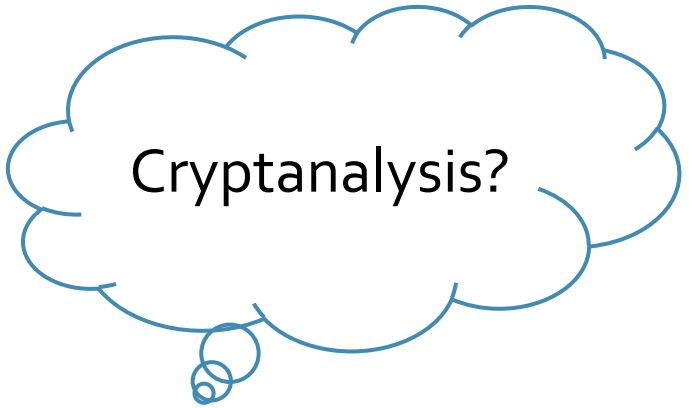
- Use any permutation of alphabetic to substitute each letter in the P.T with its corresponding (Mapping)

a	D
b	Q
c	E
d	P
e	R
f	S
g	F
h	T
i	A
j	W
k	X
l	U
m	G
n	O
o	B
p	V
q	H
r	N
s	C
t	M
u	I
v	Z
w	L
x	Y
y	J
z	K

2) Monoalphabetic Example

- P.T = computer, Key →
- C.T?

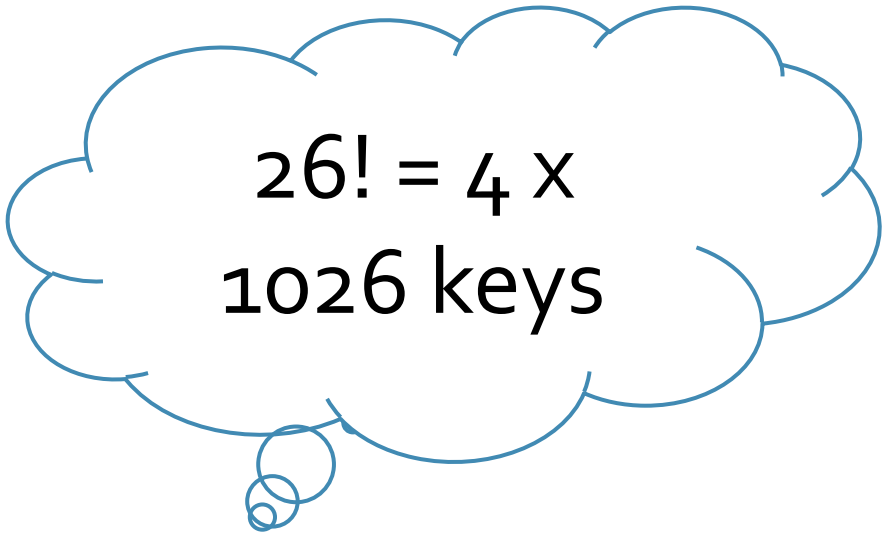


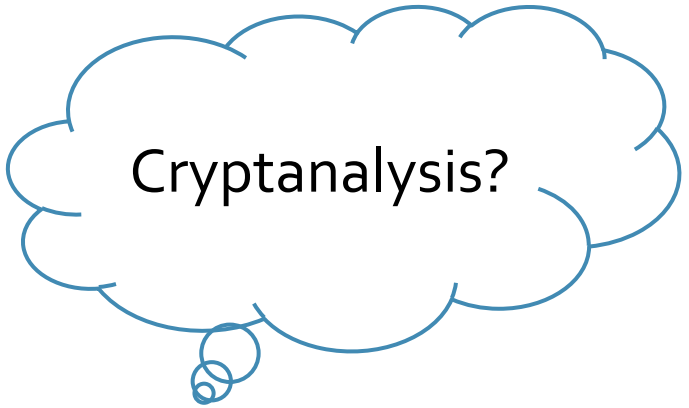


a	D
b	Q
c	E
d	P
e	R
f	S
g	F
h	T
i	A
j	W
k	X
l	U
m	G
n	O
o	B
p	V
q	H
r	N
s	C
t	M
u	I
v	Z
w	L
x	Y
y	J
z	K

2) Monoalphabetic Example

- P.T = computer, Key →
- C.T?





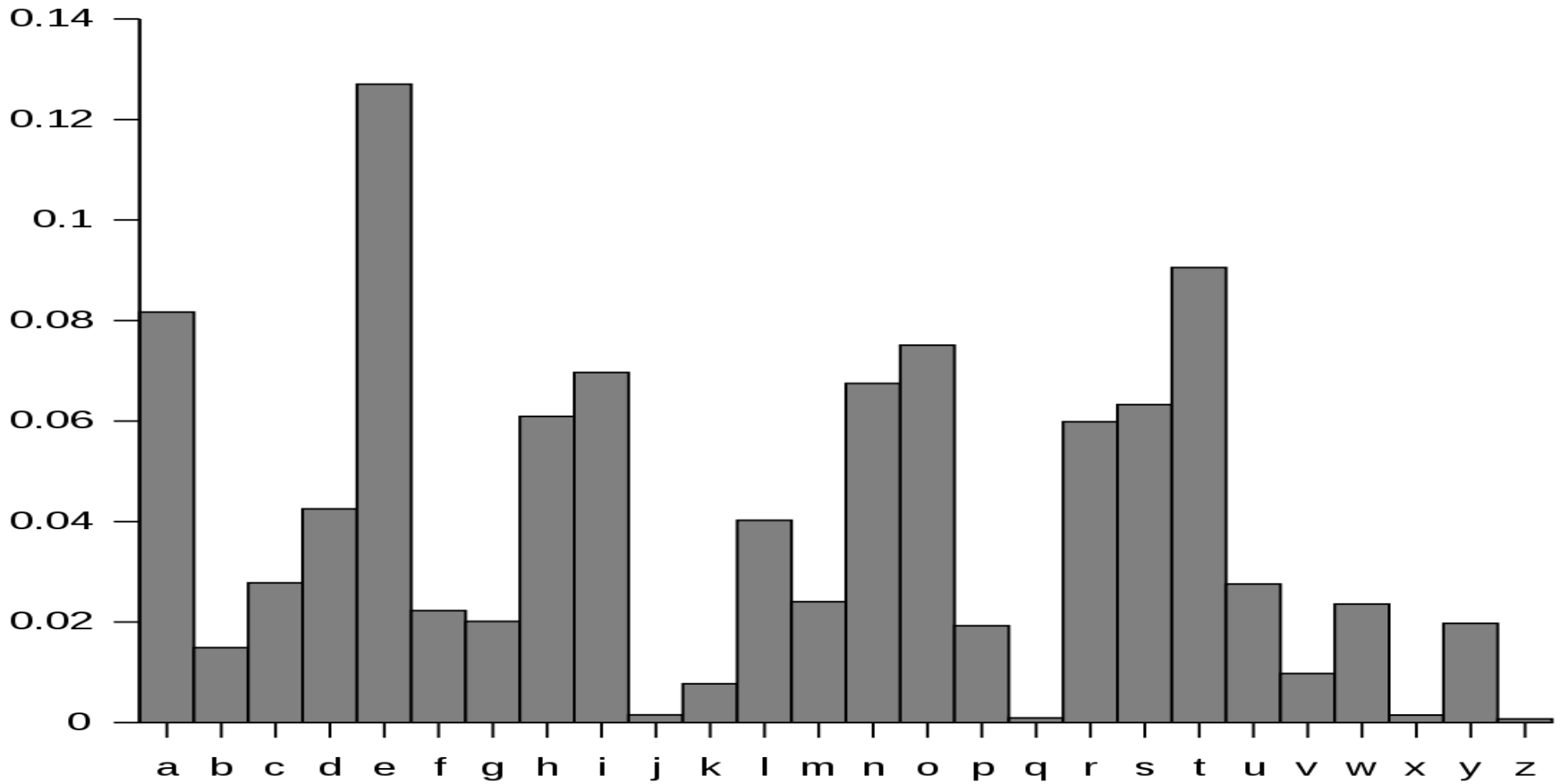
a	D
b	Q
c	E
d	P
e	R
f	S
g	F
h	T
i	A
j	W
k	X
l	U
m	G
n	O
o	B
p	V
q	H
r	N
s	C
t	M
u	I
v	Z
w	L
x	Y
y	J
z	K

2) Monoalphabetic Example

- P.T = computer, Key →
- C.T = EBGVIMRN

a	D
b	Q
c	E
d	P
e	R
f	S
g	F
h	T
i	A
j	W
k	X
l	U
m	G
n	O
o	B
p	V
q	H
r	N
s	C
t	M
u	I
v	Z
w	L
x	Y
y	J
z	K

Cryptanalysis



Cryptanalysis

Guess P & Z are e and t

Guess ZW is "th" and hence ZWP is "the"

Given ciphertext:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZVUEPHZHMDZ
SHZOWSFPAPPDTSVPQUZWYMXUZUHSXEPYEPOPDZSZUFPOMBZWPFUPZH
MDJUDTMOHMQ

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

Cryptanalysis

- Plain Text:

it was disclosed yesterday that several informal but direct contacts have been made with political representatives of the viet cong in moscow

Key concept

- Monoalphabetic substitution ciphers **do not change** relative letter frequencies.

2)Monoalphabetic Cipher

- To increase the immunity of monoalphabetic cipher
  1. **Using homophones**

a	D,-,?
b	Q,2,T
c	E,7,&
.	.
.	.
.	.

  2. **Polyalphabetic**

a	D	O
b	Q	L
c	E	V
.	.	.
.	.	.
.	.	.

  3. **Higher order substitution**

     Block cipher

aa	DS
ab	CG
ac	HH
.	.
.	.
.	.

3) Polyalphabetic

- All of Polyalphabetic cipher versions are based on using vigenere tableau.

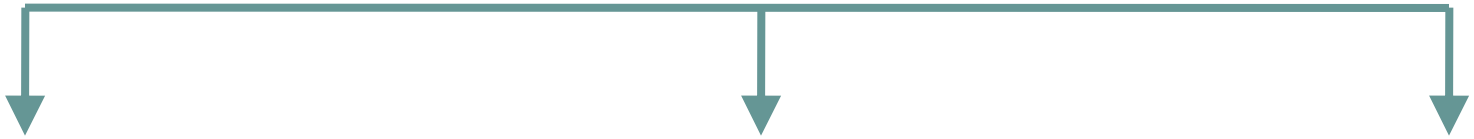
Table 2.3 The Modern Vigenère Tableau

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
b	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
c	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
d	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
e	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
f	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
g	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
h	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
i	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
j	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
k	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
l	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
m	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
n	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
o	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
p	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
r	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
s	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
t	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
u	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
v	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
w	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
x	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

3) Polyalphabetic

- All of Polyalphabetic cipher versions are based on using vigenere tableau.
- Steps:
  - Given a key word generate the key stream
  - Substitute each P. T letter with a C.T letter based on the corresponding key letter using vigenere tableau
  - Generally, **Ci = (Pi + Ki)% 26**

3) Polyalphabetic



Repeating key

Repeat the keyword till we get a key stream of length as same as the P.T length

Autokey

Append the **P.T** with the keyword to get a key stream of length as same as the P.T length

One time pad

The keyword is of length **as same as** the P.T length so the keyword is the key stream

3) Polyalphabetic: Repeating Key

- Keyword = hello
- P.T = computer
- Key-stream =hellohel
- C.T?

3) Polyalphabetic: Repeating Key

- Keyword = hello
- P.T = computer
- Key-stream =hellohel
- C.T = JSXAIAIC

3) Polyalphabetic: Repeating Key

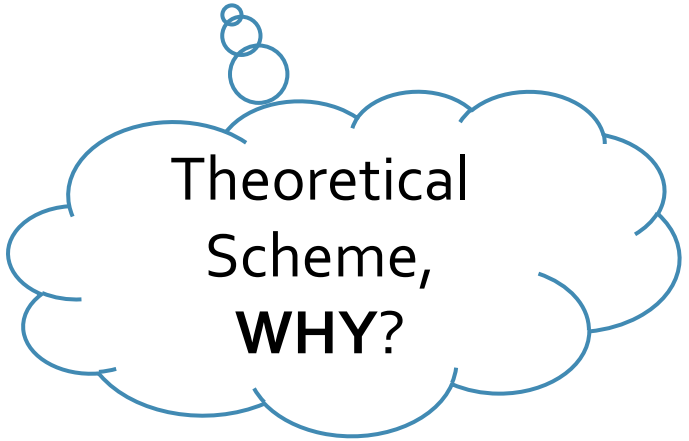
- Keyword = hello
- P.T = computer
- Key-stream =hellohel

3) Polyalphabetic: Auto-Key

- Keyword = hello
- P.T = computer
- Key-stream =hellocom
- C.T = JSXAIVSD

3) Polyalphabetic: One-time-pad

- Keyword = hilmotry
- P.T = computer




Thank
you!